



Correlation and Regression · Mark Scheme

Statistics Chapter 4: Correlation and Regression. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Scatter diagrams and correlation

- 1 (a) Positive correlation (strong) B1; (b) no (or very weak) correlation B1.
- 2 Explanatory: h , response: s B1; the number of hours of revision may affect the score, not the other way round B1.
- 3 Correlation does not imply causation B1; both are likely to be affected by a third variable, e.g. hot weather B1; more people buy ice cream and more people swim on hot days B1.
- 4 (a) The point (30, 37) B1; e.g. the runner was injured or new to running B1. (b) Positive correlation: older runners tend to take longer to run 5 km B1.
- 5(a) Weak positive: days with more sunshine tend to be slightly warmer B1 B1.
- 5(b) Sunshine s ; the amount of sunshine may affect temperature B1.
- 5(c) 18 hours is outside the range of the data (extrapolation) B1; the correlation is only weak B1.
- 5(d) Correlation does not show causation B1; data are only from June 2015 at one place, so cannot be applied to every month B1.

Section B – PMCC

- 6 $0.9 \rightarrow Q$ B1; $-0.2 \rightarrow R$ B1; $-0.85 \rightarrow P$ B1.
- 7 $S_{xx} = 190 - 150 = 40$ B1; $S_{yy} = 436 - 384 = 52$, $S_{xy} = 268 - 240 = 28$ B1; $r = \frac{28}{\sqrt{40 \times 52}}$ M1; $= 0.614$ A1.
- 8 (a) 0.73 B1; (b) -0.73 B1; adding/subtracting constants and multiplying by positive constants do not change r ; multiplying y by -1 reverses the sign B1.
- 9 $S_{xx} = 10$ B1; $S_{yy} = 34$, $S_{xy} = 17$ B1; $r = \frac{17}{\sqrt{340}}$ M1; $= 0.922$ A1.
- 10(a) $S_{xx} = 340 - 288 = 52$ B1; $S_{yy} = 13400 - 12800 = 600$ B1; $S_{xy} = 2040 - 1920 = 120$ B1.
- 10(b) $r = \frac{120}{\sqrt{52 \times 600}}$ M1; $= 0.679$ A1.
- 10(c) Moderate positive correlation: students who spend more hours on the course tend to score higher B1.
- 10(d) 0.679 B1; multiplying by a positive constant is a linear coding, which does not change r B1.

Section C – Linear regression

- 11 (a) Each extra hour adds £15 to the cost B1; (b) a fixed charge of £32 B1; (c) $y = 9.2$ (£92) B1.
- 12 $b = \frac{28}{40} = 0.7$ M1 A1; $a = 8 - 0.7 \times 5 = 4.5$, $y = 4.5 + 0.7x$ A1.
- 13 $b = \frac{17}{10} = 1.7$ M1; $a = 6 - 1.7 \times 3 = 0.9$ A1; $y = 0.9 + 1.7x$ A1.
- 14 (a) $y = 32$ B1, reliable: interpolation B1. (b) $y = 60$, unreliable: extrapolation, as 60 is outside 10–40 B1.
- 15(a) $S_{xx} = 310 - 250 = 60$ B1; $S_{xy} = 360 - \frac{50 \times 85}{10} = -65$ B1; $b = -1.083$ M1; $a = 8.5 + 1.083 \times 5 = 13.9$, $y = 13.9 - 1.08x$ A1.
- 15(b) The value decreases by about £1080 per year of age B1.
- 15(c) $y = 13.917 - 1.0833 \times 4 = 9.58$ M1; about £9580 A1.



15(d) $y = 13.917 - 16.25 = -2.33$, a negative value $B1$; impossible, since 15 years is well outside the data (extrapolation) $B1$.

Section D – Non-linear models

16 (a) $\log y$ against $\log x$ $B1$; (b) $\log y$ against x $B1$.

17 $\log y = \log a + n \log x$ $M1$; $n = 1.5$ $A1$; $a = 10^{0.48} = 3.02$, $y = 3.02x^{1.5}$ $A1$.

18 $y = 10^{0.2} \times (10^{0.3})^x$ $M1$; $k = 1.58$ $A1$; $b = 2.00$ $A1$.

19 (a) Gradient $= \frac{1.9 - 0.7}{4} = 0.3$, intercept 0.4 $M1$; $\log y = 0.4 + 0.3x \Rightarrow y = 10^{0.4}(10^{0.3})^x$ $M1$; $k = 2.51$, $b = 2.00$ $A1$.
(b) $\log y = 1.3$, $y = 20.0$ $B1$.

20(a) $N = 10^{2.1+0.15t} = 10^{2.1} \times (10^{0.15})^t$ $M1$ $M1$; $k = 125.9$ $A1$; $b = 1.413$ $A1$.

20(b) The number of bacteria is multiplied by 1.413 each hour $B1$, i.e. it grows by about 41% per hour $B1$.

20(c) $\log N = 3.6$ $M1$; $N \approx 3980$ $A1$; unreliable: $t = 10$ is outside the data range ($0 \leq t \leq 6$), and growth may slow as food runs out $B1$.

Roar-level challenge

21 $\frac{y-200}{10} = 1.2 + 0.4(x-50)$ $M1$ $A1$; $y - 200 = 12 + 4x - 200$ $M1$; $y = 12 + 4x$ $A1$; $r = 0.82$, since linear coding does not change r $B1$.